PLSC 30600

Week 7: Instrumental variables and regression discontinuity

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Roadmap

- Instrumental variables (IV): encouragement designs, Wald estimator, LATE, weak instruments
- Regression discontinuity (RD): sharp vs. fuzzy, bandwidth sensitivity, diagnostics (next)

Why instrumental variables?

In many settings, treatment receipt D_i is not randomized.

- Some units select into/out of treatment based on unobservables.
- If so, comparisons by D_i generally mix causation with selection.
- IV uses a variable Z_i that shifts D_i for some units *as good as randomly*.

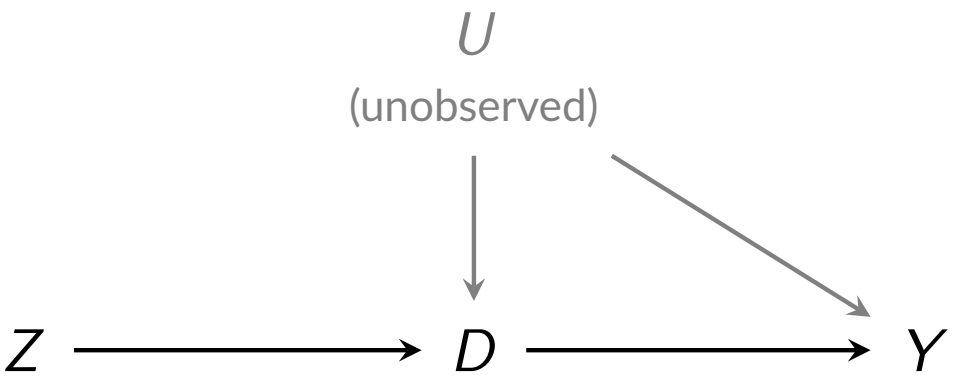
Encouragement design: assignment vs. receipt

- $Z_i \in \{0, 1\}$: randomized encouragement / assignment
- $D_i \in \{0, 1\}$: treatment actually received
- Y_i: outcome

Key idea: Z is the design feature we trust; D is endogenous.

Canonical IV DAG

- U: unobserved confounder of D and Y
- Z: instrument / encouragement
- Exclusion: no direct arrow $Z \to Y$
- Random assignment: $Z \perp\!\!\!\perp U$



Potential outcomes notation

- Define
  - Treatment potential outcomes: $Y_i(1), Y_i(0)$
  - Potential treatment receipt under encouragement: $D_i(1), D_i(0)$

- **Assume**: SUTVA
  - Observed receipt: $D_i = D_i(1)Z_i + D_i(0)(1 - Z_i)$
  - Observed outcome: $Y_i = Y_i(1)D_i + Y_i(0)(1 - D_i)$

Imbens and Angrist (1994): define counterfactuals for *receipt* $D_i(z)$ in addition to $Y_i(d)$.

Condition 0 (Exclusion): $Y_i(d, z) = Y_i(d)$ for all d, z. The instrument does not affect the outcome except through the treatment. (*Exclusion* is already built into our outcome notation here: we write $Y_i(1), Y_i(0)$ rather than $Y_i(d, z)$, so Z affects Y only through receipt D.)

Existence of instruments

Condition 1 (Instrument independence + relevance): Let Z be a random variable such that:

i. $\big(Y_i(0), Y_i(1), D_i(1), D_i(0)\big)$ is jointly independent of Z_i (as-good-as-random Z), and

ii. $\mathrm{E}\left[D_i | Z_i = z\right]$ is not constant in z. (Non-trivial first stage)

I.e., there exists a variable Z that shifts treatment receipt D for some units, and which is as good as randomly assigned.

Monotonicity

Condition 2 (Monotonicity):

- Either $D_i(1) \geq D_i(0)$ for all i, or $D_i(1) \leq D_i(0)$ for all i.

We will focus on no defiers: $D_i(1) \geq D_i(0)$ for all i.

Compliance types (principal strata)

Define types by $\big(D_i(0), D_i(1)\big)$:

Type	$D_i(0)$	$D_i(1)$
Never-taker	0	0
Always-taker	1	1
Complier	0	1
Defier	1	0

Only compliers (and defiers) change D when Z changes.

Angrist & Imbens: formal LATE identification

Theorem 1 (binary Z special case):

$$\begin{aligned} LATE_{1,0} &\equiv \mathrm{E}\left[Y(1) - Y(0) \mid D(1) - D(0) = 1\right] \\ &= \frac{\mathrm{E}\left[Y \mid Z = 1\right] - \mathrm{E}\left[Y \mid Z = 0\right]}{\mathrm{E}\left[D \mid Z = 1\right] - \mathrm{E}\left[D \mid Z = 0\right]}. \end{aligned}$$

Fixed effects / “movers” logic—we learn about effects for units whose treatment status changes.

Step 1: first stage = share of compliers

$$\begin{aligned} ITT_D &= \mathrm{E}\left[D \mid Z = 1\right] - \mathrm{E}\left[D \mid Z = 0\right] \\ &= \mathrm{E}\left[D(1) - D(0)\right]. \end{aligned}$$

By compliance type,

$$D(1) - D(0) = \begin{cases} 1 & \text{complier} \\ -1 & \text{defier} \\ 0 & \text{always / never} \end{cases}$$

So

$$ITT_D = \Pr[\text{complier}] - \Pr[\text{defier}].$$

Under monotonicity (no defiers): $ITT_D = \Pr[\text{complier}]$.

Step 2: reduced form = complier effect $\times$ share of compliers

$$\begin{aligned} ITT_Y &= \mathrm{E}\left[Y \mid Z = 1\right] - \mathrm{E}\left[Y \mid Z = 0\right] \\ &= \mathrm{E}\left[Y(D(1)) - Y(D(0))\right]. \end{aligned}$$

Using exclusion (so outcomes only depend on D) and compliance types:

$$Y(D(1)) - Y(D(0)) = \begin{cases} Y(1) - Y(0) & \text{complier} \\ Y(0) - Y(1) & \text{defier} \\ 0 & \text{always / never} \end{cases}$$

So

$$\begin{aligned} ITT_Y = \Pr[\text{complier}] \cdot \mathrm{E}\left[Y(1) - Y(0) \mid \text{complier}\right] \\ - \Pr[\text{defier}] \cdot \mathrm{E}\left[Y(1) - Y(0) \mid \text{defier}\right]. \end{aligned}$$

Under monotonicity: $ITT_Y = \Pr[\text{complier}] \cdot \mathrm{E}\left[Y(1) - Y(0) \mid \text{complier}\right].$

Wald = LATE

Combine the previous two slides:

$$\begin{aligned}\hat{\tau}_{Wald} &\equiv \frac{ITT_Y}{ITT_D} \\ &= \frac{\Pr[\text{complier}] \cdot \mathrm{E}\left[Y(1) - Y(0) \mid \text{complier}\right]}{\Pr[\text{complier}]} \\ &= \mathrm{E}\left[Y(1) - Y(0) \mid \text{complier}\right].\end{aligned}$$

Interpretation: the Wald estimator recovers the average treatment effect for the subpopulation whose treatment receipt is shifted by the instrument.

Motivation: randomized field experiment with one-sided noncompliance

Canonical example: Gerber and Green (2000): a get-out-the-vote field experiment where canvassers are randomly assigned to attempt contact.

- $Z_i \in \{0, 1\}$: randomly assigned *attempt* (assigned to canvassing).
- $D_i \in \{0, 1\}$: actual contact / receipt (some assigned units are not reached).
- One-sided noncompliance: by design, $Z_i = 0 \Rightarrow D_i = 0$ (controls are not contacted).
- Outcome Y_i: turnout.

In this setting, $\widehat{ITT}_Y = \bar{Y}_{Z=1} - \bar{Y}_{Z=0}$ is a **design-based** causal effect of assignment, and randomization inference (Week 6) is a natural way to test the sharp null of no assignment effect.

Design-based inference: only assignment is random

Week 6 perspective: treat the *potential outcomes* as fixed features of the finite population; the only randomness comes from the assignment mechanism for Z.

- Define potential outcomes under encouragement assignment: $Y_i(z)$ and $D_i(z)$, for $z \in \{0, 1\}$.

- Causal estimands (finite-population ITTs):

$$ITT_Y \equiv \bar{Y}(1) - \bar{Y}(0), \qquad ITT_D \equiv \bar{D}(1) - \bar{D}(0).$$

- Under complete randomization of Z, the difference-in-means estimators $\widehat{ITT}_Y = \bar{Y}_{Z=1} - \bar{Y}_{Z=0}$ and $\widehat{ITT}_D = \bar{D}_{Z=1} - \bar{D}_{Z=0}$ have a **design-based** sampling distribution.

Randomization inference for the reduced form / first stage

Test the sharp null of *no assignment effect* using Fisher randomization inference.

- Reduced form null: $Y_i(1) = Y_i(0)$ for all i. First stage null: $D_i(1) = D_i(0)$ for all i.
- Pick a statistic.
- Re-draw Z according to the *actual* assignment mechanism (e.g., all permutations with fixed n_1).
- The randomization distribution gives an exact p-value (finite-sample, design-based).

A non-randomized IV example: Angrist–Krueger (1991)

Angrist & Krueger (1991) instrument years of education with quarter of birth. Angrist and Krueger (1991)

- Here, Z is *not* literally randomized by a known experimenter.

- The identifying claim is an **"as-if random"** story: conditional on cohort and other controls, quarter of birth shifts schooling (relevance) but has no direct effect on wages (exclusion).

So what does "uncertainty" mean here?

- The usual 2SLS standard error is **sampling-based** (superpopulation) or **large-sample** uncertainty: if we repeatedly sampled new datasets from the same population, $\hat{\tau}_{2SLS}$ would vary.

Superpopulation inference: repeated sampling

Superpopulation view: (Y_i, D_i, Z_i, X_i) are i.i.d. draws from a population.

- The estimand is a population quantity (e.g., the β that solves a population moment condition).

- Inference is about the sampling distribution of $\hat{\beta}$ across hypothetical repeated samples.

- Large-n tools (LLN/CLT) give an approximation:

$$\sqrt{n}\,(\hat{\beta} - \beta) \;\dot{\sim}\; N(0, V),$$

so we report $SE(\hat{\beta}) \approx \sqrt{\hat{V}/n}$.

Superpopulation standard errors: what they do (and do not) capture

- A 2SLS standard error captures **sampling uncertainty** (and sometimes weak dependence assumptions), not uncertainty about whether the IV assumptions are true.

- In Angrist–Krueger, the key uncertainties are:
  - is quarter of birth "as-if random" after controls?
  - does quarter of birth affect wages only through schooling?

- Those are not solved by smaller standard errors; they require design arguments, robustness checks, and alternative specifications.

See Aronow and Miller (2019) for the separation between design-based logic and superpopulation (large-sample) approximations.

A worked Wald example (intuition)

	$Z=1$	$Z=0$
$E[Y \mid Z]$	8	5
$E[D \mid Z]$	0.75	0.25

$$\Rightarrow \qquad ITT_Y = 3, \;\; ITT_D = 0.5, \;\; \hat{\tau}_{Wald} = 6.$$

Read this as:

- encouragement increases the outcome by 3 on average
- but it only changes treatment receipt for 50% of units
- so the effect for those whose D changes must be larger.

When does the story break?

- If **defiers** exist, then $ITT_D = \Pr[\mathrm{C}] - \Pr[\mathrm{F}]$ and ITT_Y mixes complier and defier effects.
- If **exclusion fails** (direct effect of Z on Y), then always/never-takers can contribute to ITT_Y but not to ITT_D.
- If the **first stage is tiny**, the ratio is noisy (weak instrument problem).

What is 2SLS? (conceptual)

Goal: estimate the causal effect of receipt D on Y when D is endogenous.

- OLS uses *all* variation in D—including variation driven by unobservables U that also affect Y.

- An instrument Z isolates variation in D induced by Z (conditional on controls X).

- 2SLS idea: replace D with the component predicted by Z (and X), then run the outcome regression of Y on $\hat{D}$ (and the same X).

$$\hat{D}_i \equiv \mathrm{E}\left[D_i \mid Z_i, X_i\right],$$
$$D_i = \hat{D}_i + \left(D_i - \hat{D}_i\right).$$

- $\hat{D}_i$: variation in D explained by (Z, X)
- $\left(D_i - \hat{D}_i\right)$: remaining (endogenous) variation

In the simple binary-Z, no-controls case, the resulting 2SLS coefficient equals the Wald ratio.

2SLS as an algorithm (binary Z)

- First stage: regress D on Z

$$D_i = \pi_0 + \pi_1 Z_i + v_i \quad \Rightarrow \quad \hat{D}_i$$

- Second stage: regress Y on $\hat{D}$

$$Y_i = \alpha + \tau \, \hat{D}_i + e_i$$

With binary Z and no controls, the 2SLS coefficient equals the Wald ratio.

2SLS with covariates (sketch)

- First stage: $D_i = \pi_0 + \pi_1 Z_i + X_i^\top \pi_2 + v_i$
- Second stage: $Y_i = \alpha + \tau \hat{D}_i + X_i^\top \beta + e_i$
- Intuition: "use the part of D predicted by Z (after partialling out X)."

IV assumptions with covariates X

- Setup: observe (Y_i, D_i, Z_i, X_i), define $Y_i(1)$, $Y_i(0)$ and $D_i(1)$, $D_i(0)$.

- **(Conditional) independence of the instrument:**

$$\big(Y_i(0), Y_i(1), D_i(1), D_i(0)\big) \perp Z_i \mid X_i.$$

- **Exclusion:** $Y_i(d, z) = Y_i(d)$ (no direct effect of Z on Y).

- **Monotonicity (no defiers):** $D_i(1) \geq D_i(0)$ for all i.

- **First stage / relevance (given X):** $\mathrm{E}\left[D_i \mid Z_i = 1, X_i\right] \neq \mathrm{E}\left[D_i \mid Z_i = 0, X_i\right]$.

If Z is truly randomized (encouragement design), the independence condition holds unconditionally; X is then mainly for precision / adjustment.

Multiple instruments (and why matrix notation)

So far: one binary encouragement Z_i.

- In practice, we often use an *instrument set*: a vector of instruments z_i (e.g., multiple encouragement measures), plus an intercept.

- With controls X_i, 2SLS typically treats the controls as instruments for themselves.

$$\mathrm{z}_i \equiv \begin{bmatrix} 1 \\ Z_i \\ X_i \end{bmatrix} \quad \text{(one instrument + controls)}$$

$$\mathrm{z}_i \equiv \begin{bmatrix} 1 \\ Z_{1i} \\ \vdots \\ Z_{Li} \\ X_i \end{bmatrix} \quad \text{(multiple instruments + controls).}$$

This is why the 2SLS formula is easiest to write using the projection onto the space spanned by the instrument matrix.

2SLS in one equation (matrix form)

Let X be the full regressor matrix in the structural equation (endogenous + exogenous), and Z the instrument matrix (all exogenous variables used as instruments).

Stacking the instrument vectors z_i gives $Z = [z_1^\top; \ldots; z_n^\top]$.

Define the projection matrix onto the instrument space:

$$P_Z \equiv Z\left(Z^\top Z\right)^{-1} Z^\top.$$

Then the 2SLS estimator is:

$$\hat{\beta}_{2SLS} = \left(X^\top P_Z X\right)^{-1} X^\top P_Z Y.$$

Equivalent view: first *project* the regressors onto $\mathrm{span}(Z)$, then run OLS in that projected space.

2SLS with controls = "partial out, then do IV"

Structural equation with controls:

$$Y_i = \tau D_i + X_i^\top \beta + \varepsilon_i.$$

Equivalent implementation (Frisch–Waugh–Lovell / "partialling out"):

- regress Y on X and keep residuals $\tilde{Y}$
- regress D on X and keep residuals $\tilde{D}$
- regress each instrument on X and keep residuals $\tilde{Z}$
- run 2SLS of $\tilde{Y}$ on $\tilde{D}$, instrumenting with $\tilde{Z}$ (same $\hat{\tau}$ as "2SLS with controls")

This makes the covariate-adjusted case look like the simple Wald/IV case again.

In practice: just include X in both stages and let software compute standard errors; don't hand-residualize unless you're careful about inference.

Toy example: OLS vs. Wald vs. 2SLS

Simulate endogeneity (an unobserved U affects both D and Y), but a randomized encouragement Z shifts D.

Result: OLS is biased, while Wald = 2SLS uses only Z-induced variation in D (targeting the complier LATE).

```
> set.seed(60637)
> n <- 2000
> Z <- rbinom(n, 1, 0.5)
> U <- rnorm(n); v <- rnorm(n); eps <- rnorm(n)
> D_star <- 0.2 + 1.0 * Z + 0.8 * U + v
> D <- as.numeric(D_star > 0)
> tau <- 2
> Y <- tau * D + U + eps
> ITT_Y <- mean(Y[Z == 1]) - mean(Y[Z == 0])
> ITT_D <- mean(D[Z == 1]) - mean(D[Z == 0])
> wald <- ITT_Y / ITT_D
> ols <- coef(lm(Y ~ D))[["D"]]
> X <- cbind(1, D); Zmat <- cbind(1, Z)
> Pz <- Zmat %*% solve(t(Zmat) %*% Zmat) %*% t(Zmat)
> beta_2sls <- solve(t(X) %*% Pz %*% X) %*% (t(X) %*% Pz %*% Y)
> tsls <- as.numeric(beta_2sls[2])
> round(c(true = tau, ols = ols, wald = wald, tsls = tsls), 3)
```

```
 true   ols  wald  tsls
2.000 2.911 1.745 1.745
```

Code: 2SLS with `ivreg`

```
> library(AER)  # provides ivreg()
> df <- data.frame(Y, D, Z)
> fit_iv <- ivreg(Y ~ D | Z, data = df)
> summary(fit_iv, diagnostics = TRUE)
Call:
ivreg(formula = Y ~ D | Z, data = df)

Residuals:
     Min       1Q   Median       3Q      Max
-5.64481 -0.99357 -0.03347  0.97968  5.16765

Coefficients:
            Estimate Std. Error t value Pr(>|t|)
(Intercept)   0.1867     0.2041   0.915     0.36
D             1.7448     0.2873   6.072 1.51e-09 ***

Diagnostic tests:
                  df1  df2 statistic  p-value
Weak instruments    1 1998    124.22  < 2e-16 ***
Wu-Hausman          1 1997     20.56 6.12e-06 ***
Sargan              0   NA        NA       NA
---
Signif. codes:  0 ‘***’ 0.001 ‘**’ 0.01 ‘*’ 0.05 ‘.’ 0.1 ‘ ’ 1

Residual standard error: 1.423 on 1998 degrees of freedom
Multiple R-Squared: 0.4242,        Adjusted R-squared: 0.4239
```

IV / 2SLS as moment-based estimation (GMM / minimum distance)

Core IV moment condition:

$$\mathrm{E}\left[\mathsf{Z}_i\left(Y_i - \mathsf{X}_i^\top \beta\right)\right] = 0.$$

Sample analog:

$$u_i(\beta) \equiv Y_i - \mathsf{X}_i^\top \beta,$$

$$\hat{g}(\beta) \equiv \frac{1}{n}\sum_{i=1}^{n} \mathsf{Z}_i\, u_i(\beta).$$

- **Just-identified**: # moments = # parameters $\Rightarrow$ solve $\hat{g}(\hat{\beta}) = 0$.
- **Overidentified**: # moments $>$ # parameters $\Rightarrow$ choose $\hat{\beta}$ to make moments "as close to 0 as possible":

$$\hat{\beta} = \arg\min_{\beta}\ \hat{g}(\beta)^\top \hat{W}\, \hat{g}(\beta).$$

This is why "IV", "2SLS", and "GMM/minimum distance" fit under one umbrella.

Weak instruments (Staiger & Stock 1997)

- If the first stage is weak, TSLS/2SLS behaves poorly:
  - finite-sample bias toward OLS
  - standard errors / test sizes can be misleading
- Diagnostics:
  - first-stage F-statistic (rule-of-thumb: $F \gtrsim 10$ as a warning flag)
  - first-stage partial R^2
- Conceptual link: weak first stage $\Rightarrow$ dividing by a small number in the Wald ratio.

(Optional) Multi-valued Z: IV as an average of local effects

Canonical IV estimand:

$$\tau_{IV}(g) = \frac{\mathrm{Cov}(Y, g(Z))}{\mathrm{Cov}(D, g(Z))}.$$

Imbens–Angrist (1994) show: with multi-valued Z, under monotonicity plus an "aligned" $g(Z)$ condition, $\tau_{IV}(g)$ is a weighted average of adjacent local effects $LATE_{z_k, z_{k-1}}$ (nonnegative weights that sum to 1).

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